

Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment Bundle Practices for Critically Ill Children: An International Survey of 161 PICUs in 18 Countries

OBJECTIVES: To evaluate current international practice in PICUs regarding components of the “Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment” (“ABCDEF”) bundle.

DESIGN: Online surveys conducted between 2017 and 2019.

SETTING: One-hundred sixty-one PICUs across the United States ($n = 82$), Canada ($n = 14$), Brazil ($n = 27$), and Europe ($n = 38$) participating in the Prevalence of Acute Rehabilitation for Kids in the PICU study.

INTERVENTIONS: None.

MEASUREMENTS AND MAIN RESULTS: Of the 161 participating PICUs, 83% were in academic teaching hospitals and 42% were in free-standing children’s hospitals. Median size was 16 beds (interquartile range, 10–24 beds). Only 15 PICUs (9%) had incorporated all six ABCDEF bundle components into routine practice. Standardized pain assessment (A) was the most common (91%), followed by family engagement (F, 88%) and routine sedation assessment (C) with validated scales (84%). Protocols for testing extubation readiness or conducting spontaneous breathing trials (B) were reported in 57%, with 34% reporting a ventilator weaning protocol. Routine delirium monitoring with a validated screening tool (D) was reported by 44% of PICUs, and 26% had a guideline, protocol, or policy for early exercise/mobility (E). Practices for spontaneous breathing trials were variable in 29% of Canadian PICUs versus greater than 50% in the other regions. Delirium monitoring was lowest in Brazilian PICUs (18%) versus greater than 40% in other regions, and family engagement was reported in 55% of European PICUs versus greater than 90% in other regions.

CONCLUSIONS: ABCDEF bundle components have been adopted with substantial variability across regions. Additional research must rigorously evaluate the efficacy of specific elements with a focus on B, D, E, and full ABCDEF bundle implementation. Implementation science is needed to facilitate an understanding of the barriers to ABCDEF implementation and sustainability with a focus on specific cultural and regional differences.

KEY WORDS: delirium; early mobilization; family engagement; intensive care units; pediatric; sedation

Erwin Ista, RN, PhD¹

Juliana Redivo, MD²

Paurav Kananur²

Karen Choong, MB, BCh, MSc,
FRCP^{3,4}

Jose Colleti Jr, MD⁵

Dale M. Needham, MD, PhD^{6,7}

Ronke Awojodu, MPH, RN²

Sapna R. Kudchadkar, MD, PhD,
FCCM^{2,7,8,9}

on behalf of the International
PARK-PICU Investigators

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The Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment (ABCDEF) bundle is the foundation of the Society of Critical Care Medicine's ICU Liberation collaborative (1, 2). This multicomponent and evidence-based six-step strategy was developed to liberate adult patients from life support initiated during acute critical illness, including facilitating mobilization out of bed and family engagement/empowerment, with the goal of decreasing ICU-acquired morbidities (3). The ABCDEF bundle represents A: Assess, prevent, and manage pain; B: Both Spontaneous Awakening Trials (SATs) and Spontaneous Breathing Trials (SBTs); C: Choice of sedation; D: Delirium assessment, prevention, and management; E: Early mobility and exercise; and F: Family engagement and empowerment. The concept of ICU Liberation gained significant attention in the adult critical care literature and has more recently engaged pediatric critical care practitioners (1, 4).

Optimization of pain and sedation management has been integral to the care of critically ill infants and children since the inception of pediatric critical care as a specialty. In 2013, an international survey of pediatric intensivists was conducted to examine sedation, sleep promotion, and delirium screening practices for mechanically ventilated children (5). In that study, less than 2% of respondents reported routine delirium screening at least once every 12 hours, and most did not report use of validated delirium screening tools. Although 70% of pediatric intensivists reported working in PICUs that used standardized sedation assessment and documentation, only 42% of those respondents used these assessments to direct sedation. More recently, delirium has been a major area of focus in the PICU, with development of validated screening tools and increased knowledge of modifiable risk factors, including benzodiazepines and physical restraints (6–8).

A strong evidence base in the adult critical care literature supports the positive clinical impact of each individual ABCDEF bundle component as well as the collective bundle (9–12). However, data to support components of this bundle in critically ill children are limited and/or recently emerging, particularly related to spontaneous breathing trials (B), choice of sedation and analgesia (C), delirium (D), and early mobility (E) (6, 13–15). Therefore, understanding current practice

of the ABCDEF bundle and its components in PICUs is relevant for identifying future research priorities. Thus, we aimed to characterize the prevalence of these ABCDEF practices among PICUs across the world.

METHODS

A survey of PICU clinical practices was sent to 161 sites that participated in the Prevalence of Acute Rehabilitation in the PICU (PARK-PICU) study in Brazil, Canada, the United States, and Europe. PARK-PICU was a 2-day point prevalence study conducted separately in three global regions to characterize clinical practices for critically ill children (<https://park.web.jhu.edu>). All surveys were completed between October 2017 and October 2019. Participating PICUs were recruited via the Pediatric Acute Lung Injury and Sepsis Investigators Network, Canadian Critical Care Trials Group, Associação de Medicina Intensiva Brasileira (Brazilian Critical Care Society), and European Society of Pediatric and Neonatal Intensive Care, along with e-mails, social media, and a dedicated website (<https://park.web.jhu.edu>). PICUs were eligible to participate if they: 1) had a separate physical space dedicated to caring for infants and children and 2) were capable of providing mechanical ventilation. The survey was administered via REDCap (the United States, Canada, Brazil) or LimeSurvey (Europe) to PARK-PICU study leads at each site, who were instructed to complete the survey in collaboration with the PICU's physicians, nurse manager, and rehabilitation team leader to ensure accuracy in responses. All participating PARK-PICU sites obtained Institutional Review Board approval prior to completing the survey.

Survey Components

Survey items were developed by an interprofessional team, including pediatric critical care physicians, nurses, respiratory therapists, and rehabilitation clinicians. The survey questions included the type of hospital (freestanding, academic), number of PICU beds, type of patients (medical-surgical, cardiac, mixed), and physical aspects of the unit (private vs shared rooms). The following sections then queried about practices within each component of the ABCDEF bundle. Specifically, survey questions asked about: 1) pain and withdrawal assessment tools and frequency (A = Assessing Pain); 2) ventilator weaning, daily

awakening trials, and extubation readiness testing (ERT) (B = Both SAT and SBT); 3) sedation assessment tools and frequency and sedation protocols (C = Choice of Sedation); 4) delirium screening tools and frequency (D = Delirium Monitoring and Management); 5) early mobilization programs/protocols and personnel and equipment resources (E = Early Mobility and Exercise); and 6) policies and practices surrounding family involvement (F = Family Engagement and Empowerment). The organizational survey was developed using available conduct and reporting guidance (16) and is available at <https://park.web.jhu.edu>.

Statistical Analysis

Descriptive statistics are summarized as counts and percentages for categorical variables and as medians and interquartile ranges (IQRs) or means and SDs, for continuous variables. Implementation rates of ABCDEF bundle components were compared across PICU characteristics, including geographic regions, using chi-square analysis or Kruskal-Wallis test. A two-tailed p value of less than 0.05 was considered statistically significant. Data were analyzed with Microsoft Excel for Mac version 16 (Microsoft, Redmond, WA) and IBM SPSS Version 25.0 (IBM, Armonk, NY).

RESULTS

PICU Characteristics

Surveys were completed by all PICUs that participated in the PARK-PICU study ($n = 161$ PICUs; 148 hospitals) (**Supplemental Fig. 1**, Supplemental Digital Content 1, <http://links.lww.com/CCM/G574> and **Supplemental Table 1**, Supplemental Digital Content 2, <http://links.lww.com/CCM/G574>). Most PICUs (83%) were in an academic teaching hospital (133/161; $p < 0.001$), and 42% (68/161; $p < 0.001$) were in a free-standing children's hospital (Supplemental Table 1, Supplemental Digital Content 2, <http://links.lww.com/CCM/G574>). The overall median number of PICU beds was 16 (IQR, 10–24) but differed significantly among the four regions ($p < 0.001$), with the United States having the highest median number of PICU beds (24; IQR, 16–30) and Brazil the lowest (10; IQR, 7–16). Approximately half of all PICUs (55%) had single-patient rooms, with significant variation between

regions, ranging from 13% (Europe) to 71% (Canada and the United States).

Assessing Pain

Most PICUs reported routinely documenting and assessing pain (91%; 147/161) (**Table 1**). The most common pain scales used were the Face, Legs, Activity, Cry, Consolability scale (71%) and the Numerical Rating Scale (53%), with significant variation between the regions. In addition to the Wong-Baker FACES scale (43%), other pain assessment tools reported included the Neonatal Infant Pain Scale ($n = 8$; 5%), the COMFORT scale ($n = 7$; 4%), and the Neonatal Pain, Agitation and Sedation Scale ($n = 6$; 4%). Similarly, the frequency of pain assessment and documentation varied substantially between the regions, ranging from every hour to every 12 hours.

Both SAT and SBT

Only 14% of PICUs (22/161) reported routine use of SATs in intubated patients; however, 53% (85/161) implemented a protocol for SBTs in the form of ERT (**Table 1**). Ventilator weaning protocols were reported by 34% of PICUs (55/161), usually conducted by respiratory therapists. Brazilian and European PICUs were more likely to have implemented an ERT protocol than were U.S. and Canadian PICUs (70% and 63% vs 48% and 21%, respectively; $p = 0.010$).

Choice of Sedation and Assessment of Withdrawal

Sedation was routinely assessed and documented in 84% of PICUs (135/161); however, assessment was significantly less common in Brazil (67%) and Canada (71%) than in Europe and the United States (89% and 90%, respectively; $p = 0.0191$) (**Table 2**). The State Behavioral Scale was most commonly used in Canada and the United States (50% and 56%), and the COMFORT behavior scale was more frequently used in Brazil and Europe (50% and 77%). About half of PICUs (51%; 82/161) reported a unit-based sedation protocol, and when a protocol was present, medications were titrated by physicians/nurse practitioners (51%), nurses only (23%), or both (26%). The majority of PICUs (80%) reported routine assessment and documentation of iatrogenic withdrawal syndrome (IWS)

TABLE 1.

PICU Practices by Region for A and B Components of the Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment Bundle

Survey Items	Brazil (N = 27), n (%)	Canada (N = 14), n (%)	Europe (N = 38), n (%)	The United States (N = 82), n (%)	Total N = 161	p ^a
Assess pain (A)						
Routine documentation and assessment of pain	21 (78)	11 (79)	37 (97)	78 (95)	147 (91)	0.006
Scale ^b						< 0.001
Numerical Rating Scale	12 (44)	5 (36)	12 (32)	57 (69)	86 (53)	
Face, Legs, Activity, Cry, Consolability scale	17 (63)	10 (71)	18 (47)	70 (85)	115 (71)	
Wong-Baker FACES	6 (22)	5 (36)	2 (5)	57 (70)	70 (44)	
Oucher	1 (4)	0	0	4 (5)	5 (3)	
Other ^b	1 (4)	3 (21)	13 (34)	17 (21)	34 (21)	
Frequency of assessment and documentation						0.002
Every 2 hr	5 (24)	0	7 (18)	35 (45)	47 (32)	
Every 4 hr	5 (24)	6 (54)	10 (26)	24 (31)	45 (30)	
Other	11 (52)	5 (46)	21 (55)	19 (24)	56 (38)	
Both spontaneous awakening and breathing trials (B)						
Routinely perform awakening trials in intubated patients						0.001
No	21 (78)	10 (72)	28 (74)	46 (56)	105 (65)	
Yes	2 (7)	2 (14)	10 (26)	8 (10)	22 (14)	
Sometimes	4 (15)	2 (14)	0	28 (34)	34 (21)	
Protocol for ERT	19 (70)	3 (21)	24 (63)	39 (48)	85 (53)	0.010
ERT can be ordered by:						< 0.001
Respiratory therapist	5 (26)	0	0	3 (8)	8 (10)	
Physician/nurse practitioner	5 (26)	2 (67)	24	16 (42)	47 (56)	
Both	9 (47)	1 (33)	0	19 (50)	29 (35)	
Ventilator weaning protocol	15 (56)	3 (21)	13 (34)	24 (29)	55 (34)	0.061
Ventilator weaning protocol driven by:						< 0.001
Respiratory therapist	1 (7)	0	0	7 (29)	8 (15)	
Physician/nurse practitioner	4 (27)	0	13 (100)	4 (17)	21 (38)	
Both	10 (67)	3 (100)	0	13 (54)	26 (47)	

ERT = extubation readiness testing.

^a χ^2 was used to compare characteristics across regions.

^bCould be used more > 1 scale, percentage could be > 100%.

(Table 2), with most using the Withdrawal Assessment Tool-1 (79%) and Sophia Observation withdrawal Symptoms (SOS; 13%) scale. The frequency of

withdrawal assessment and documentation varied widely, with every 12 hours being the most common (41%).

TABLE 2.

PICU Practices by Region for C and D Components of the Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment Bundle

Survey Items	Brazil (N = 27), n (%)	Canada (N = 14), n (%)	Europe (N = 38), n (%)	The United States (N = 82), n (%)	Total N = 161	p ^a
Choice of sedation and analgesia (C)						
Routine documentation and assessment of sedation	18 (67)	10 (71)	34 (90)	73 (89)	135 (84)	0.0191
Use of sedation scale ^b						< 0.001
State Behavioral Scale	0	5 (50)	3 (9)	41 (56)	49 (36)	
Richmond Agitation Sedation Scale	9 (50)	1 (10)	1 (3)	18 (25)	29 (22)	
COMFORT (behavior) scale	9 (50)	1 (10)	26 (77)	3 (4)	39 (29)	
Ramsay	0	1 (10)	0	1 (1)	2 (2)	
Other ^c	0	2 (20)	4 (12)	10 (14)	16 (12)	
Frequency of sedation assessment and documentation						0.021
Every 2 hr	2 (11)	0	9 (36)	27 (37)	38 (30)	
Every 4 hr	5 (28)	7 (70)	7 (28)	24 (33)	43 (34)	
Other	11 (62)	3 (30)	9 (36)	22 (30)	45 (36)	
Sedation protocol	14 (52)	4 (29)	24 (63)	40 (49)	82 (51)	0.155
Protocol titrated by:						< 0.001
Nursing only	0	1 (25)	2 (8)	16 (40)	19 (23)	
Physician/nurse practitioner only	13 (93)	1 (25)	22 (92)	6 (15)	42 (51)	
Both	1 (7)	2 (50)	0	18 (45)	21 (26)	
Routine assessment and documentation of withdrawal	9 (33)	14 (100)	30 (79)	75 (92)	128 (80)	< 0.001
Scales						< 0.001
Withdrawal Assessment Tool-1	9 (100)	12 (86)	12 (40)	68 (92)	101 (79)	
Sophia Observation withdrawal Scale	0	0	14 (47)	2 (3)	16 (13)	
Other ^c	0	2 (14)	5 (16)	4 (5)	11 (9)	
Frequency of withdrawal assessment and documentation						0.464
Every 4 hr	1 (11)	4 (29)	8 (26)	22 (29)	35 (27)	
Every 12 hr	4 (44)	8 (57)	15 (48)	26 (35)	53 (41)	
Other	4 (44)	2 (14)	8 (26)	27 (36)	41 (32)	
Delirium monitoring and management (D)						
Routine delirium screening and documentation	6 (22)	6 (43)	16 (42)	42 (51)	70 (44)	0.072
Tools						0.004
Cornell Assessment Pediatric Delirium	3 (50)	6 (100)	7 (44)	35 (83)	51 (74)	
Pediatric Confusion Assessment Method for ICU	3 (50)	0	1 (6)	8 (19)	12 (17)	
Preschool Confusion Assessment Method for ICU	1	0	0	5 (12)	6 (9)	
Sophia Observation Withdrawal Symptoms-Pediatric Delirium scale	0	0	7 (44)	0	7 (10)	
Other	0	0	1 (6%)	3 (7)	4 (6)	
Delirium assessment and documentation every 12 hr	3 (50)	6 (100)	8 (50)	38 (90)	55 (74)	< 0.001

^a χ^2 was used to compare characteristics across regions.

^bCould be used more > 1 scale, percentage could be > 100%.

^cOther sedation assessment tools reported included the Pediatric Sedation Scale, the Penn State Children's Hospital Sedation Algorithm ($n = 3$), the University of Michigan Sedation Scale ($n = 2$), the Neonatal Pain, Agitation and Sedation Scale ($n = 2$), the Pasero tool ($n = 2$), and the Riker Sedation-Agitation Scale ($n = 1$).

Delirium Monitoring and Management

Nearly half of all PICUs (44%; 70/161) reported routine delirium screening and documentation (Table 2). Seventy-four percent of PICUs that screened for delirium routinely did so once per shift (every 12 hr). Most PICUs used the Cornell Assessment of Pediatric Delirium (74%); however, screening tools differed among the four regions (Table 2). Half of the Brazilian PICUs also used the pediatric Confusion Assessment Method for the ICU, and in Europe, the SOS-Pediatric Delirium scale was used in half of PICUs.

Early Mobility and Exercise

Only 26% of PICUs reported an early mobilization or rehabilitation protocol, most commonly in the form of an order set (56%) or unit-based guideline (24%) (Table 3). In fact, early mobility was the least common practice of all six bundle components. Only 10% of PICUs with an early mobilization protocol reported a written PICU policy for it. Brazilian (89%), Canadian (57%), and European (61%) PICUs were more likely to have a dedicated physical therapist than were U.S. PICUs (18%; $p < 0.001$). PICUs were less likely to have an occupational therapist or speech language pathologist, defined as greater than or equal to one full-time equivalent, with no differences between the four regions (Table 3).

Family Engagement and Empowerment

Most PICUs (88%) allowed at least one family member at the bedside 24 hours per day; however, this practice was least common in European PICUs (55%). Furthermore, 98% of PICUs (86/88) with private rooms allowed family members at the bedside 24 hours per day (Table 3). Nearly, all Canadian PICUs (100%) and U.S. PICUs (94%) invited families to participate in bedside rounds compared with only 40% in Europe and 33% in Brazil ($p < 0.001$). Only 23% of PICUs reported providing families with formal documents or training to encourage engagement in their child's care.

Combination of ABCDEF Bundle Components

PICUs were grouped according to combinations of ABCDEF bundle components used as part of routine practice (Fig. 1). All ABCDEF bundle components

were implemented in practice in 9% of the PICUs (15/161) and varied across the regions ($p = 0.097$): Canadian (0%), European (3%), Brazilian (7%), and the United States (15%). Furthermore, the combination of bundle components used as part of clinical practices was diverse (Fig. 2). Combinations of "ABCF," "ACDF," and "ACF" were most frequently reported (12–16%).

DISCUSSION

We found that implementation of the ABCDEF bundle components varies widely across Canada, the United States, Europe, and Brazil, with full bundle implementation achieved in only 9% of participating PICUs. Analgesia and sedation assessment and family engagement are more consistently implemented with important regional differences. The last similar survey of pediatric intensivists internationally in 2013 focused only on sedation, sleep, and delirium practices (5). As such, this unit-based survey provides an update of current practice on a multinational level with inclusion of extubation readiness trials, early mobility, and family engagement on a multinational level. Universal goals in PICU care include reducing the occurrence of preventable harm, including pain, distress, IWS, delirium, and ICU-acquired weakness. These preventable harms may be mitigated by integration of the B, D, and E elements of the bundle; however, it is these very components that are the least commonly practiced in the surveyed PICUs.

Adult ICU studies have shown that full ABCDEF bundle performance is associated with improved survival, duration of mechanical ventilation, delirium and coma, physical restraint use, ICU readmission, and discharge disposition (17). However, application of the ABCDEF bundle in pediatrics poses numerous unique challenges. Although the evidence for each component in the PICU continues to grow, the burgeoning movement to combat postintensive care syndrome is driving more PICUs to review their practices to improve both functional outcomes for PICU survivors and the patient-family experience (18–20).

We observed substantial variability in family presence across regions. Family engagement and family-centered care are not novel concepts in pediatrics, as they are core values of clinical teams that care for children. Nevertheless, nearly half of European PICUs do

TABLE 3.

PICU Practices by Region for E and F Components of the Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment Bundle

Survey Items	Brazil (N = 27), n (%)	Canada (N = 14), n (%)	Europe (N = 38), n (%)	The United States (N = 82), n (%)	Total (N = 161), n (%)	p ^a
Early mobility and exercise (E)						
Early mobilization/rehabilitation protocol	12 (44)	2 (14)	5 (13)	22 (27)	41 (26)	0.027
The protocol consists of:						
Order set	6 (50)	1 (50)	5 (100)	11 (50)	23 (56)	
Published hospital policy	6 (50)	0	0	4 (18)	10 (24)	
Guideline	3 (25)	2 (100)	0	12 (55)	17 (41)	
Dedicated therapy staff in the PICU (≥ 1 full-time equivalent)						
Physical therapist ^a	24 (89)	8 (57)	23 (61)	15 (18)	70 (44)	< 0.001
Occupational therapist	3 (11)	4 (29)	7 (18)	13 (16)	27 (17)	0.544
Speech language pathologist	5 (19)	1 (7)	0	10 (12)	16 (13)	0.549
Family engagement and empowerment (F)						
A family member is allowed to be at the bedside 24 hr/d	25 (93)	14 (100)	21 (55)	82 (100)	142 (88)	< 0.001
Private rooms, n	15	10	5	58	88	
Family allowed at bedside in private rooms	14 (93)	10 (100)	4 (80)	58 (100)	86 (98)	0.019
Families are invited to participate in bedside rounds	9 (33)	14 (100)	15 (40)	77 (94)	115 (71)	< 0.001
Families are given documents or training to encourage engagement in their child's care	3 (11)	1 (7)	20 (53)	13 (16)	37 (23)	< 0.001

^aχ² was used to compare characteristics across regions.

not allow for family presence at the bedside 24 hours a day, particularly if they have multipatient rooms. We believe this is most likely due to practical barriers than specific cultural differences between Europe and other countries. Furthermore, we found that only 23% of all surveyed PICUs provide information or documentation to encourage family engagement in the care of their child. Family-centered care is substantially broader than open visiting policies (21, 22). Therefore, education about family engagement (both written and interactive discussion with PICU teams) should include how families can participate in PICU rounds and patient care (e.g., washing, comforting, or mobilizing) and how they can support other family members (e.g., siblings) for coping with a child's PICU admission (23). As identified in PARK-PICU United States, Europe, and Canada, family presence and involvement

are independently associated with out-of-bed mobility of PICU patients (24–26).

Our findings for the analgesia and sedation components are consistent with prior literature, with striking variability in tools used for both domains. This variability is likely described by the multitude of pain and sedation tools validated for use in critically ill infants and children and recommended in international guidelines (27). Thus, it is a wake-up call to the pediatric critical care community to consider streamlining which tools are used globally. Assessment of IWS, as we observed, is more common PICU practice with only two validated tools (28, 29); however, it is rarely studied in adults and thus not part of the ABCDEF bundle (30). The presence of IWS is stressful for the critically ill child and their parents (31) and leads to prolonged hospital stay and increased healthcare costs (32); thus, addition of

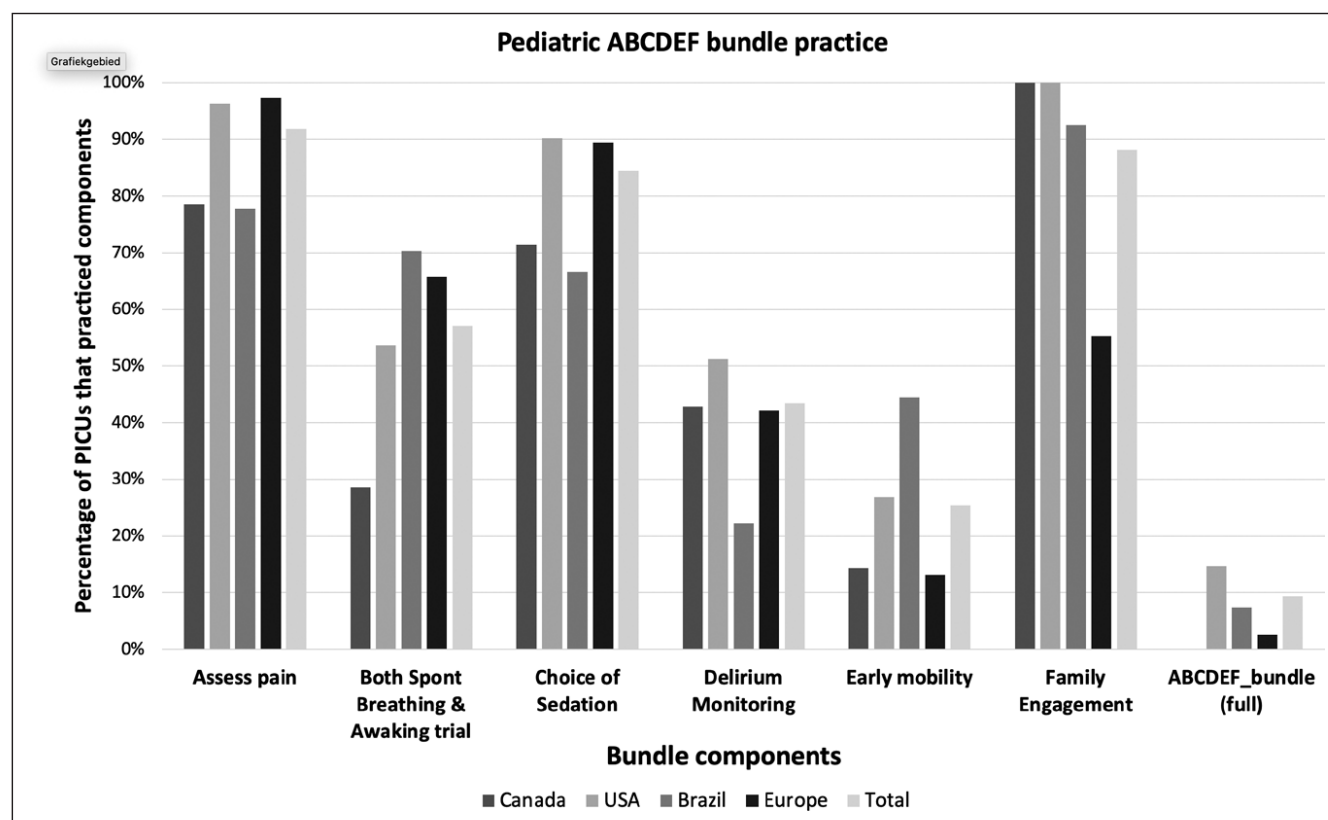


Figure 1. Pediatric Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment (ABCDEF) bundle practice in PICUs by region. Number of PICUs: Canada, $n = 14$; United States, $n = 82$; Brazil, $n = 27$; Europe, $n = 38$; total $n = 161$.

IWS to the ABCDEF bundle must be considered particularly for pediatrics. We also found that sedation protocols were implemented in daily practice in half of PICUs. Although a large multicenter clinical found no reduction in duration of mechanical ventilation or PICU stay with protocolized sedation (14), nurse-driven sedation protocols have continued to show feasibility and potential benefits for reducing sedative and analgesic exposure and consequently prevalence of IWS (33, 34). Integrating IWS assessment into the routine use of the ABCDEF bundle provides a bundled approach to prevention of IWS with goal-directed sedation and analgesia and early mobility.

Half of PICUs surveyed routinely practice ERT, whereas only 14% perform SATs. The B element of the ABCDEF bundle focuses on setting a time each day to pause sedative medications, orient the patient to time and day, and conduct an SBT in an effort to liberate the patient from the ventilator. SBTs are mostly used in combination with SAT, also known as daily sedation interruption. In adult ICU patients, SBT and SAT synergistically help decrease mechanical ventilation

days, hospital lengths of stay, and delirium (35). Both, SBT and SAT have limited evidence in PICU populations (36, 37). However, a large cluster randomized stepped wedge trial in 18 PICUs is underway in the United Kingdom to determine the effectiveness of protocolized sedation and SBT on duration of mechanical ventilation (38).

Delirium is highly prevalent in critically ill children with validated tools for screening (7, 8, 39, 40). Although delirium screening has increased since 2013 thanks to heightened attention in the adult literature and validation of more pediatric-specific tools (7, 8, 39, 40), screening occurs in only 20–40% of the surveyed PICUs in this study, this is in line with other surveys (25–29%) (5, 41, 42). Screening is the first major step in delirium prevention and treatment using both nonpharmacologic and, when indicated, pharmacologic approaches (43). Reducing or avoiding deliriogenic medications including benzodiazepines is advantageous and recommended in adult ICU guidelines (44). Thus, application of delirium screening as part of the ABCDEF bundle, not alone, is critical given

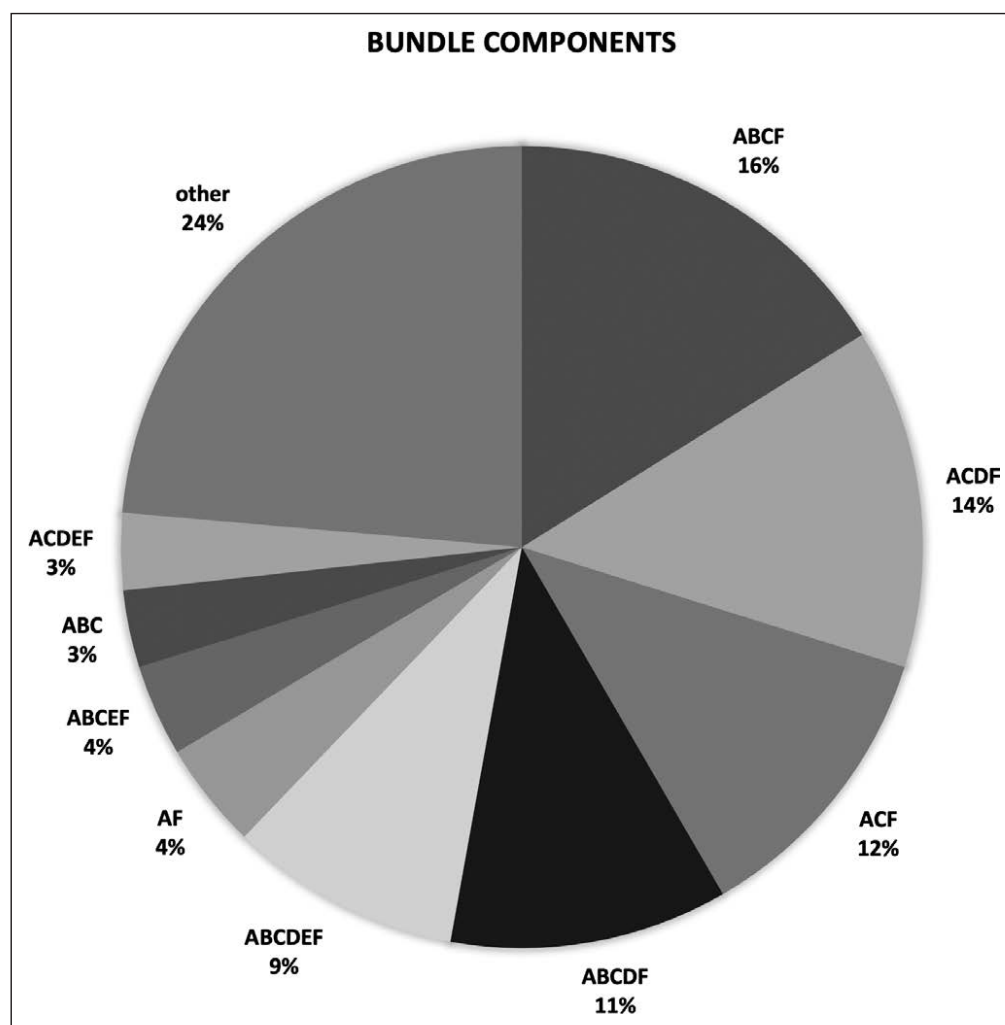


Figure 2. Combinations of Assessing Pain, Both Spontaneous Awakening and Breathing Trials, Choice of Sedation, Delirium Monitoring/Management, Early Exercise/Mobility, and Family Engagement/Empowerment (ABCDEF) bundle components in practice. Only combinations of bundle components practiced in five or more PICUs are presented in the pie chart. The section called “other” consists of all bundle combinations practiced in fewer than five PICUs.

the bundle’s aim to minimize oversedation with medications which may contribute to delirium and optimize family engagement and early mobility which are important tools in preventing delirium. Effectiveness of the bundle should be based on outcomes most important to patients and families including health-related quality of life, sleep, and physical mobility (45).

Insufficient or inconclusive evidence for an intervention can be an important barrier for large-scale implementation. In addition to creating evidence for pediatric ABCDEF bundle use, barriers and facilitators for its implementation must also be explored. In a recent systematic review, 107 barriers to ABCDEF bundle implementation in adult ICUs were identified and catalogued into four relevant domains: 1)

individual patient (e.g., hemodynamic instability, safety concerns); 2) clinician related (e.g., lack of knowledge, agreement, and confidence with protocols or guidelines, perceived workload); 3) protocol related (e.g., unavailability of protocols and criteria, lack of evidence); and 4) ICU context (e.g., culture, lack of leadership, and interprofessional communication) (46). Similar barriers have been acknowledged regarding delirium screening or early mobilization in the PICU (47–49). In fact, a recent survey of clinicians in Chinese PICUs revealed that about half of the respondents were not aware of the ABCDE bundle strategy (42). Interprofessional PICU teams can use these domains and related barriers as a guide to develop and select tailored implementation strategies that take into consideration their own PICU’s specific

context and barriers. A systematic approach will likely improve implementation and fidelity of the ABCDEF bundle and, ultimately, improve patient care and outcomes.

Our study has several limitations. First, we surveyed only PICUs included in the PARK-PICU study. Generalizability to other PICUs and geographic regions may be limited. However, based on other international surveys (5, 41), we expect that results may be similar. Second, the surveys were conducted at different time points over a 2-year period. However, it is unlikely that important temporal trends occurred that would affect our findings during this period. Third, variability in rehabilitation staffing and practices across regions may limit translation of findings.

For example, U.S. PICUs have fewer dedicated PTs than do the other countries. However, in Brazil and Europe, PTs perform respiratory/pulmonary therapy as well as physical therapy/rehabilitation (50). Finally, as a cross-sectional survey of unit-based practice, we could not characterize the association between ABCDEF bundle practice and patient-based outcomes including mortality or duration of mechanical ventilation or PICU stay; prospective studies are ongoing to delineate this relationship (51).

CONCLUSIONS

In this international survey of 161 PICUs from 18 countries, we found that ABCDEF bundle components have been adopted to varying degrees in Canada, the United States, Brazil, and Europe, with substantial differences between geographic regions. There is an urgency for additional research on the efficacy of specific components including Breathing, Delirium, and Early Mobility and full ABCDEF bundle implementation in critically ill children. It is also important to understand barriers for implementation of the ABCDEF bundle and determine whether cultural and regional differences affect implementation.

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TN); Meredith F. Bone, Sara VandenBranden (Ann & Robert H. Lurie Children's Hospital of Chicago, Chicago, IL); Gokul Kris Bysani (Medical City Children's Hospital, Dallas, TX); Maddie Chrisman, Ericka L. Fink (UPMC Children's Hospital of Pittsburgh, Pittsburgh, PA); LeeAnn Christie (Dell Children's Medical Center, Austin, TX); Jean Christopher (Akron Children's Hospital, Akron, OH); Christina Cifra, Weerapong Lilitwat (The University of Iowa, Iowa City, IA); David S. Cooper, Alicia Rice (Cincinnati Children's Hospital Medical Center CICU, Cincinnati, OH); Allison S. Cowl (Connecticut Children's Medical Center, Hartford, CT); Jason W. Custer (University of Maryland Children's Hospital, Baltimore, MD); Melissa G. Chung (Nationwide Children's Hospital PICU, Columbus, OH); Danielle Van Damme, Kristen A. Smith (C.S. Mott Children's Hospital, Ann Arbor, MI); Erin Bennett, Rebecca Dixon (University of Utah Primary Children's Hospital, Salt Lake City, UT); Molly V. Dorfman, Ashley Mancini (Valley Children's Hospital, Madera, CA); Sharon P. Dial (Cohen Children's Medical Center, New Hyde Park, NY); Jane L. Di Gennaro, Leslie A. Dervan (Seattle Children's Hospital, Seattle, WA); Lesley Doughty, Laura Benken (Cincinnati Children's Hospital Medical Center PICU, Cincinnati, OH); Mark C. Dugan, Judith Ben Ari (Children's Hospital of Nevada at UMC, Las Vegas, NV); Melanie Cooper Flaigle, Vianne Smith (Providence Sacred Heart Children's Hospital, Spokane, WA); Shira J. Gertz (St Barnabas Medical Center, Livingston, NJ); Katherine Gregersen, Shamel A. Abd-Allah (Loma Linda University Health, Loma Linda, CA); Justin Hamrick, Katherine Irby (University of Arkansas PICU & CVICU, Little Rock, AR); Jodi Herbsman, Yasir M. Al-Qaqaa (NYU Langone Medical Center, New York, NY); John Holcroft (University of California Davis, Davis, CA); Erin Hulfish, Kathleen Culver (Stony Brook Children's Hospital, Stony Brook, NY); Susan Hupp, Andrea DeMonbrun (UTSW Children's Medical Center CVICU, Dallas, TX); Kelechi Iheagwara, Shelli Lavigne-Sims (Our Lady of the Lake Children's Hospital, Baton Rouge, LA); Christine Joyce, Chani Traube (Weill Cornell Medical College, New York, NY); Pradip Kamat, Cheryl Stone (Egleston Children's Hospital at Emory University PICU, Atlanta, GA); Sameer S. Kamath, Melissa Harward (Duke Children's Hospital & Health Center PICU, Durham, NC);

Priscilla Kaszubski, Joanne Daguanno (St Joseph's Children's Hospital, Paterson, NJ); Robert P. Kavanagh, Debbie Spear (Penn State Hershey Children's Hospital, Hershey, PA); Yu Kawai, Karen Fryer (Mayo Clinic, Rochester, MN); Bree Kramer (John R. Oishei Children's Hospital, Buffalo, NY); Erin M. Kreml, Brian T. Burrows (Phoenix Children's Hospital PICU and CVICU, Phoenix, AZ); Andrew W. Kiragu (Hennepin County Medical Center, Minneapolis, MN); Truc M. Le, Stacey R. Williams (Monroe Carell Jr. Children's Hospital at Vanderbilt, PCICU, Nashville, TN); John C. Lin, Amanda Florin (St. Louis Children's Hospital, Saint Louis, MO); Peter M. Lockett, Tammy Robertson (UTSW Children's Medical Center PICU, Dallas, TX); Vanessa N. Madrigal, Ashleigh B. Harlow (Children's National Health System, Washington, DC); Barry Markovitz, Fernando Beltramo (Children's Hospital Los Angeles CICU, Los Angeles, CA); Michael C. McCrory (Wake Forest University Baptist Medical Center, Winston Salem, NC); Robin L. McKinney (Hasbro Children's Hospital, Providence, RI); Maryam Y. Naim (Children's Hospital of Philadelphia CICU, Philadelphia, PA); Asha G. Nair, Ravi Thiagarajan (Boston Children's Hospital CICU, Boston, MA); Shilpa Narayan, Kathleen Murkowski (Children's Hospital of Wisconsin/Medical College of Wisconsin, Milwaukee, WI); Keshava Murthy Narayana Gowda, Jhoclay See (Cleveland Clinic, Cleveland, OH); Pooja A. Nawathe (Cedars-Sinai Medical Center, Los Angeles, CA); William E. Novotny, Cynthia Keel (Vidant Medical Center, Greenville, NC); Peter Oishi, Neelima Marupudi (UCSF Benioff Children's Hospital, San Francisco, CA); Laura Ortmann (Children's Hospital & Medical Center, Omaha, NE); A. M. Iqbal O'Meara, Nikki Miller Ferguson (Children's Hospital of Richmond at VCU, Richmond, VA); Megan E. Peters (University of Wisconsin, Madison, WI); Neethi Pinto, Allison Kniola (The University of Chicago, Chicago, IL); Courtney M. Rowan, Jill Mazurczyk (Riley Hospital for Children at Indiana University, Indianapolis, IN); Shilpa Shah, Sage Lachman (Children's Hospital Los Angeles CTICU, Los Angeles, CA); Marcy N. Singleton, Sholeen T. Nett (Dartmouth Hitchcock Medical Center, Lebanon, NH); Michael C. Spaeder, Jenna V. Zschaebitz (University of Virginia, Charlottesville, VA); Thomas Spentzas (Le Bonheur Children's Hospital PICU, Memphis, TN); Sue S. Sreedhar (Johns Hopkins All Children's Hospital, St.

Petersburg, FL); Katherine M. Steffen, Michelle Chen (Lucile Packard Children's Hospital Stanford, Palo Alto, CA); Anne Stormorken, Allison Blatz (Rainbow Babies & Children's Hospital, Cleveland, OH); Sachin D. Tadphale (Le Bonheur Children's Hospital CVICU, Memphis, TN); Robert C. Tasker, John F. Griffin (Boston Children's Hospital MSICU, Boston, MA); Tammy L. Uhl, Melissa Harward (Duke Children's Hospital & Health Center CICU, Durham, NC); Karen H. Walson, Cynthia Bates (Scottish Rite Children's Hospital at Emory University, Atlanta, GA); Christopher M. Watson, Mary Lynn Sheram (Children's Hospital of Georgia at Augusta University, Augusta, GA); Cydni N. Williams, Aileen Kirby (OHSU Doernbecher Children's Hospital, Portland, OR); Michael Wolf, Kellet Lowry (Egleston Children's Hospital at Emory University CICU, Atlanta, GA); Heather A. Wolfe (Children's Hospital of Philadelphia PICU, Philadelphia, PA); and Andrew R. Yates, Brian Beckman (Nationwide Children's Hospital CTICU, Columbus, OH). **Europe:** PARK-PICU EU Collaborators: Paul C. Ritson MCSP, Alder Hey Children's NHS Foundation Trust, Liverpool, United Kingdom; Filippia Nikolaou, MD, Children Hospital "P & A" Aglaia Kyriakou, Athens, Greece; Marjorie de Neef, RN, MSc, AMC-Emma Children's Hospital, Amsterdam, The Netherlands; Martin Kneyber, MD, PhD, Beatrix Children's Hospital, Groningen, The Netherlands; Barnaby R. Scholefield, MBBS, MRCPCH, PhD, Birmingham Women & Children's Hospital, Paediatric ICU, Birmingham, United Kingdom; Kate Penny-Thomas, RM, Birmingham Women & Children's Hospital, Birmingham, United Kingdom, UK; Christina Linton, Bristol Royal Hospital for Children, Bristol, United Kingdom; Reinis Balmaks, MD, Children's Clinical University Hospital, Riga, Latvia; Matthias Richter, MD, Universitätsklinikum Carl Gustav Carus Klinik für Kinder- und Jugendmedizin, Dresden, Germany; Fabrizio Chiusolo, MD, Children's Hospital Bambino Gesù—PICU, Rome, Italy; Corrado Cecchetti, MD, Children's Hospital Bambino Gesù - Emergency Department PICU, Rome, Italy; Marco Roberti, MSc, RN, Children's Hospital Bambino Gesù -CICU, Rome, Italy; Michela Di Furia, MSc, RN, Children's Hospital Bambino Gesù—Palidoro, Italy; Alicja Bartkowska-Śniatkowska, MD, PhD, Department of Pediatric Anesthesiology and Intensive Therapy, Poznań University of Medical Sciences,

Poznań, Poland; Chantal Grandjean, RN, MSc, Lausanne University Hospital (CHUV), Lausanne, Switzerland; Bettina Nygaard, RN, PhD, Copenhagen University Hospital, Rigshospitalet, Copenhagen, Denmark; Yolanda Lopez, MD, Cruces University Hospital, Bilbao, Spain; Tolga Koroglu, MD, Dokuz Eylül University, Izmir, Turkey; Tolga Besci, MD, Dokuz Eylül University, Izmir, Turkey; Simona Calza, RN, PhD, IRCCS Istituto Giannina Gaslini, Genova, Italy; Rachel S. Agbeko, MD, PhD, Great North Children's Hospital, Newcastle upon Tyne, United Kingdom; Emma Borrows, MD, Great Ormond Street Hospital, London, United Kingdom; Nathalie Bochaton, MSc, Geneva University Hospital (HUG), Geneva, Switzerland; Orsola Gawronski, RN, MS, PhD, Healthcare Professional Development Unit, Bambino Gesù Children's Hospital, Rome, Italy; Anne-Sylvie Ramelet, PhD, Institute of Higher Education and Research in Healthcare, University of Lausanne and Lausanne University Hospital, Lausanne, Switzerland; Janet Mattsson, RN, PhD, Karolinska, Stockholm, Sweden; Anne Ksellmann, MD, Kinderherzzentrum Sankt Augustin, St Augustin, Germany; Barbara Hero, RN, Children's University Hospital Cologne, Köln, Germany; Jowita Rosada-Kurasinska, Department of Pediatric Anesthesiology and Intensive Therapy, Poznań University of Medical Sciences, Poznań, Poland; Magdalena Świder, MD, Kliniczny Szpital Wojewódzki No 2 w Rzeszowie, Rzeszów, Poland; Irene Harth, RN, PICU, Universitätsmedizin Mainz, Zentrum für Kinder- und Jugendmedizin, Mainz, Germany; Amabile Bonaldi, RN, Ospedale Maggiore, Verona, Italy; Cristina Giugni, MD, Meyer Children's Hospital, Florence, Italy; Siva Oruganti, MD, Noah's Ark Children's Hospital for Wales, Cardiff, United Kingdom; Joseph C. Manning, PhD, RN, Nottingham Children's Hospital & Neonatology, Nottingham University Hospitals NHS Trust, Nottingham, United Kingdom; Simon Gates, PT, MSc, Nottingham Children's Hospital and Neonatology, Nottingham University Hospitals NHS Trust, Nottingham, United Kingdom; Hazel Smith, RM, PhD, Children's Health Ireland at Crumlin, Dublin, Ireland; Annelies van Zwol, MD, PhD, Radboudumc-Amila Children's Hospital, Nijmegen, The Netherlands; Jenna Hills, PT, Royal Hospital for Children, Glasgow, Scotland; Johanna Conroy, PT, Royal Manchester Children's Hospital, Manchester,

United Kingdom; Mark Bebbington, MD, Royal Stoke University Hospital, Stoke-on-Tren, United Kingdom; Felix Neunhoffer, MD, PhD, Department of Pediatric Cardiology, Pulmonology and Pediatric Intensive Care Medicine, University Children's Hospital Tübingen, Tübingen, Germany; Jozef De Dooy, MD, PhD, UZA Antwerp, Belgium. **Canada:** Karen Choong, David J. Zorko (McMaster Children's Hospital, McMaster University, Hamilton, ON, Canada); Laurence Ducharme-Crevier (CHU Sainte-Justine, University of Montreal, Montreal, QC, Canada); Patricia S. Fontela (Montreal Children's Hospital, McGill University, Montreal, QC, Canada); Laurie A. Lee (Alberta Children's Hospital, University of Calgary, Calgary, AB, Canada); Anne-Marie Guerguerian (The Hospital for Sick Children, University of Toronto, Toronto, ON, Canada); Gonzalo Garcia Guerra (Stollery Children's Hospital, University of Alberta, Edmonton, AB, Canada); Kristina Krmpotic (IWK Health Centre, Dalhousie University, Halifax, NS, Canada); Brianna McKelvie (London Health Sciences Centre, Western University, London, ON, Canada); Kusum Menon (Children's Hospital of Eastern Ontario, University of Ottawa, Ottawa, ON, Canada); Srinivas Murthy (BC Children's Hospital, University of British Columbia, Vancouver, BC, Canada); Anupam Sehgal (Kingston Health Science Center, Queen's University, Kingston, ON, Canada); Matthew J. Weis (CHU de Quebec, Université Laval Faculté de Médecine, Québec, QC, Canada). **Brazil:** Nelson Kazunobu Horigoshi, MD (Hospital Infantil Sabará, São Paulo, Brazil); Graziela de Araújo Costa, MD (Hospital Sírio-Libanês, São Paulo, Brazil); Taísa Roberta Ramos de Castilho, MD (Hospital Anália Franco Rede D'Or São Luiz and Hospital Beneficência Portuguesa, São Paulo, Brazil); Paula Peres Domingues Peron, MD (Instituto de Criança, Hospital das Clínicas da Universidade de São Paulo, São Paulo, Brazil); Walter Perez Scaranto, MD (Hospital Municipal Carmino Caricchio—Tatuapé, São Paulo, Brazil); Daniela Nasu Monteiro Medeiros, MD (Hospital Municipal Dr Moyses Deutch—M'Boi Mirim, São Paulo, Brazil); Toshio Matsumoto, MD (Hospital Municipal Infantil Menino Jesus, São Paul, Brazil); Carlos Gustavo de Almeida, MD (Hospital Assunção Rede D'Or São Luiz, SB do Campo, Brazil); Felipe Rezende Caino de Oliveira, MD (Instituto de Oncologia Pediátrica—GRAACC, São Paulo, Brazil) Marcelo Barciela Brandão, MD (Universidade de

Campinas—UNICAMP, Campinas, Brazil) Fernanda Lima-Setta, MD (Instituto Fernandes Figueira—FIOCRUZ, Rio de Janeiro, Brazil); Arnaldo Prata-Barbosa, MD (Hospital Copa D'Or, Hospital Quinta D'Or, Hospital Caxias D'Or, Hospital Rios D'Or, Hospital Oeste D'Or, Hospital Real D'Or, Rio de Janeiro, Brazil); Glaciele Nascimento Xavier, RT (Instituto de Cardiologia do Distrito Federal, Brasília, Brazil); Livia Barbosa de Andrade, RT (Hospital Esperança, Recife, Brazil); Agda Ultra de Aguiar, RT (Hospital de Base do Distrito Federal, Brasília, Brazil); Marcos Paulo Galdino Coutinho, RT (Hospital Otávio de Freitas, Recife, Brazil); Roberta Esteves Viera de Castro, MD (Hospital Universitário Pedro Ernesto—UERJ, Rio de Janeiro, Brazil); Glazia André Landy, RT (Instituto de Tratamento do Câncer Infantil—ITACI, São Paulo, Brazil); Suzana Lopes Bonfim Balaniuc, RT (Hospital Universitário Maria Aparecida Pedrossian, Campo Grande, Brazil); Ricardo Silveira Yamaguchi, MD (Hospital da Luz, São Paulo, Brazil).

- 1 Department of Pediatric Surgery, Intensive Care Unit, Erasmus MC-Sophia Children's Hospital, University Medical Center Rotterdam, Rotterdam, The Netherlands.
- 2 Department of Anesthesiology and Critical Care Medicine, Johns Hopkins University School of Medicine, Baltimore, MD.
- 3 Department of Pediatrics, McMaster University, Hamilton, ON, Canada.
- 4 Department of Health Research Methods, Evidence, and Impact, McMaster University, Hamilton, ON, Canada.
- 5 Department of Pediatrics, Hospital Assunção – Rede D'Or São Luiz, SB do Campo, São Paulo, Brazil.
- 6 Division of Pulmonary and Critical Care Medicine, Department of Medicine, Johns Hopkins University School of Medicine, Baltimore, MD.
- 7 Outcomes after Critical Illness and Surgery Group, Johns Hopkins University, Baltimore, MD.
- 8 Department of Physical Medicine & Rehabilitation, Johns Hopkins University School of Medicine, Baltimore, MD.
- 9 Department of Pediatrics, Charlotte R. Bloomberg Children's Center, Johns Hopkins University School of Medicine, Baltimore, MD.

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For information regarding this article, E-mail: w.ista@erasmusmc.nl

REFERENCES

1. Ely EW: The ABCDEF bundle: Science and philosophy of how ICU liberation serves patients and families. *Crit Care Med* 2017; 45:321–330
2. Barr J, Fraser GL, Puntillo K, et al; American College of Critical Care Medicine: Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med* 2013; 41:263–306
3. Pandharipande P, Banerjee A, McGrane S, et al: Liberation and animation for ventilated ICU patients: The ABCDE bundle for the back-end of critical care. *Crit Care* 2010; 14:157
4. Morandi A, Piva S, Ely EW, et al: Worldwide survey of the “assessing pain, both spontaneous awakening and breathing trials, choice of drugs, delirium monitoring/management, early exercise/mobility, and family empowerment” (ABCDE) bundle. *Crit Care Med* 2017; 45:e1111–e1122
5. Kudchadkar SR, Yaster M, Punjabi NM: Sedation, sleep promotion, and delirium screening practices in the care of mechanically ventilated children: A wake-up call for the pediatric critical care community*. *Crit Care Med* 2014; 42:1592–1600
6. Traube C, Silver G, Reeder RW, et al: Delirium in critically ill children: An international point prevalence study. *Crit Care Med* 2017; 45:584–590
7. Traube C, Silver G, Kearney J, et al: Cornell assessment of pediatric delirium: A valid, rapid, observational tool for screening delirium in the PICU*. *Crit Care Med* 2014; 42:656–663
8. Smith HA, Boyd J, Fuchs DC, et al: Diagnosing delirium in critically ill children: Validity and reliability of the pediatric confusion assessment method for the intensive care unit. *Crit Care Med* 2011; 39:150–157
9. Balas MC, Vasilevskis EE, Olsen KM, et al: Effectiveness and safety of the awakening and breathing coordination, delirium monitoring/management, and early exercise/mobility bundle. *Crit Care Med* 2014; 42:1024–1036
10. Pandharipande PP, Pun BT, Herr DL, et al: Effect of sedation with dexmedetomidine vs lorazepam on acute brain dysfunction in mechanically ventilated patients: The MENDS randomized controlled trial. *JAMA* 2007; 298:2644–2653
11. Tipping CJ, Harrold M, Holland A, et al: The effects of active mobilisation and rehabilitation in ICU on mortality and function: A systematic review. *Intensive Care Med* 2017; 43:171–183
12. Trogrlić Z, van der Jagt M, Bakker J, et al: A systematic review of implementation strategies for assessment, prevention, and

- management of ICU delirium and their effect on clinical outcomes. *Crit Care* 2015; 19:157
13. Abu-Sultaneh S, Hole AJ, Tori AJ, et al: An interprofessional quality improvement initiative to standardize pediatric extubation readiness assessment. *Pediatr Crit Care Med* 2017; 18:e463–e471
 14. Curley MA, Wypij D, Watson RS, et al; RESTORE Study Investigators and the Pediatric Acute Lung Injury and Sepsis Investigators Network: Protocolized sedation vs usual care in pediatric patients mechanically ventilated for acute respiratory failure: A randomized clinical trial. *JAMA* 2015; 313:379–389
 15. Wiecek B, Ascenzi J, Kim Y, et al: PICU Up!: Impact of a quality improvement intervention to promote early mobilization in critically ill children. *Pediatr Crit Care Med* 2016; 17:e559–e566
 16. Kelley K, Clark B, Brown V, et al: Good practice in the conduct and reporting of survey research. *Int J Qual Health Care* 2003; 15:261–266
 17. Pun BT, Balas MC, Barnes-Daly MA, et al: Caring for critically ill patients with the ABCDEF bundle: Results of the ICU liberation collaborative in over 15,000 adults. *Crit Care Med* 2019; 47:3–14
 18. Manning JC, Pinto NP, Rennick JE, et al: Conceptualizing post intensive care syndrome in children-the PICS-p framework. *Pediatr Crit Care Med* 2018; 19:298–300
 19. Needham DM, Davidson J, Cohen H, et al: Improving long-term outcomes after discharge from intensive care unit: Report from a stakeholders' conference. *Crit Care Med* 2012; 40:502–509
 20. Herrup EA, Wiecek B, Kudchadkar SR: Characteristics of post-intensive care syndrome in survivors of pediatric critical illness: A systematic review. *World J Crit Care Med* 2017; 6:124–134
 21. Hill C, Knafl KA, Santacroce SJ: Family-centered care from the perspective of parents of children cared for in a pediatric intensive care unit: An integrative review. *J Pediatr Nurs* 2018; 41:22–33
 22. Richards CA, Starks H, O'Connor MR, et al: Elements of family-centered care in the pediatric intensive care unit: An integrative review. *J Hosp Palliat Nurs* 2017; 19:238–246
 23. Herrup EA, Wiecek B, Kudchadkar SR: Feasibility and perceptions of PICU diaries. *Pediatr Crit Care Med* 2019; 20:e83–e90
 24. Ista E, Scholefield BR, Manning JC, et al; EU PARK-PICU Collaborators: Mobilization practices in critically ill children: A European point prevalence study (EU PARK-PICU). *Crit Care* 2020; 24:368
 25. Kudchadkar SR, Nelliott A, Awojodu R, et al; Prevalence of Acute Rehabilitation for Kids in the PICU (PARK-PICU) Investigators and the Pediatric Acute Lung Injury and Sepsis Investigators (PALISI) Network: Physical rehabilitation in critically ill children: A multicenter point prevalence study in the United States. *Crit Care Med* 2020; 48:634–644
 26. Choong K, Zorko DJ, Awojodu R, et al: Prevalence of acute rehabilitation for kids in the PICU: A Canadian multicenter point prevalence study. *Pediatr Crit Care Med* 2021; 22:181–193
 27. Harris J, Ramelet AS, van Dijk M, et al: Clinical recommendations for pain, sedation, withdrawal and delirium assessment in critically ill infants and children: An ESPNIC position statement for healthcare professionals. *Intensive Care Med* 2016; 42:972–986
 28. Franck LS, Scoppettuolo LA, Wypij D, et al: Validity and generalizability of the withdrawal assessment tool-1 (WAT-1) for monitoring iatrogenic withdrawal syndrome in pediatric patients. *Pain* 2012; 153:142–148
 29. Ista E, de Hoog M, Tibboel D, et al: Psychometric evaluation of the sophia observation withdrawal symptoms scale in critically ill children. *Pediatr Crit Care Med* 2013; 14:761–769
 30. Arroyo-Novoa CM, Figueroa-Ramos MI, Balas M, et al: Opioid and benzodiazepine withdrawal syndromes in trauma ICU patients: A prospective exploratory study. *Crit Care Explor* 2020; 2:e0089
 31. Craske J, Carter B, Jarman I, et al: Parent's experiences of their child's withdrawal syndrome: A driver for reciprocal nurse-parent partnership in withdrawal assessment. *Intensive Crit Care Nurs* 2019; 50:71–78
 32. Ista E, van Dijk M, Gischler S, et al: Weaning of opioids and benzodiazepines at home after critical illness in infants: A cost-effective approach. *J Opioid Manag* 2010; 6:55–62
 33. Neunhoffer F, Kumpf M, Renk H, et al: Nurse-driven pediatric analgesia and sedation protocol reduces withdrawal symptoms in critically ill medical pediatric patients. *Paediatr Anaesth* 2015; 25:786–794
 34. Gaillard-Le Roux B, Liet JM, Bourgoin P, et al: Implementation of a nurse-driven sedation protocol in a PICU decreases daily doses of Midazolam. *Pediatr Crit Care Med* 2017; 18:e9–e17
 35. Girard TD, Kress JP, Fuchs BD, et al: Efficacy and safety of a paired sedation and ventilator weaning protocol for mechanically ventilated patients in intensive care (awakening and breathing controlled trial): A randomised controlled trial. *Lancet* 2008; 371:126–134
 36. Vet NJ, de Wildt SN, Verlaet CW, et al: A randomized controlled trial of daily sedation interruption in critically ill children. *Intensive Care Med* 2016; 42:233–244
 37. Ferreira FV, Sugo EK, Aragon DC, et al: Spontaneous breathing trial for prediction of extubation success in pediatric patients following congenital heart surgery: A randomized controlled trial. *Pediatr Crit Care Med* 2019; 20:940–946
 38. Blackwood B, Agus A, Boyle R, et al; Paediatric Intensive Care Society Study Group (PICS-SG): Sedation AND Weaning In Children (SANDWICH): Protocol for a cluster randomised stepped wedge trial. *BMJ Open* 2019; 9:e031630
 39. Ista E, van Beusekom B, van Rosmalen J, et al: Validation of the SOS-PD scale for assessment of pediatric delirium: A multicenter study. *Crit Care* 2018; 22:309
 40. Smith HA, Gangopadhyay M, Goben CM, et al: The preschool confusion assessment method for the ICU: Valid and reliable delirium monitoring for critically ill infants and children. *Crit Care Med* 2016; 44:592–600
 41. Staveski SL, Pickler RH, Lin L, et al: Management of pediatric delirium in pediatric cardiac intensive care patients: An international survey of current practices. *Pediatr Crit Care Med* 2018; 19:538–543
 42. Huang X, Lei L, Zhang S, et al: Implementation of the "awakening and breathing trials, choice of drugs, delirium management, and early exercise/mobility" bundle in the pediatric intensive care unit of tertiary hospitals in southwestern

- China: A cross-sectional survey. *J Int Med Res* 2021; 49:300060520987770
43. van den Boogaard M, Slooter AJC, Brüggemann RJM, et al; REDUCE Study Investigators: Effect of haloperidol on survival among critically ill adults with a high risk of delirium: The REDUCE randomized clinical trial. *JAMA* 2018; 319:680–690
 44. Devlin JW, Skrobik Y, Gélinas C, et al: Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med* 2018; 46:e825–e873
 45. Fink EL, Maddux AB, Pinto N, et al; Pediatric Outcomes Studies after PICU (POST-PICU) Investigators of the Pediatric Acute Lung Injury and Sepsis Investigators (PALISI) Network and the Eunice Kennedy Shriver National Institute of Child Health and Human Development Collaborative Pediatric Critical Care Research Network (CPCCRN): A core outcome set for pediatric critical care. *Crit Care Med* 2020; 48:1819–1828
 46. Costa DK, White MR, Ginier E, et al: Identifying barriers to delivering the awakening and breathing coordination, delirium, and early exercise/mobility bundle to minimize adverse outcomes for mechanically ventilated patients: A systematic review. *Chest* 2017; 152:304–311
 47. Joyce CL, Taipe C, Sobin B, et al: Provider beliefs regarding early mobilization in the pediatric intensive care unit. *J Pediatr Nurs* 2018; 38:15–19
 48. Simone S, Edwards S, Lardieri A, et al: Implementation of an ICU bundle: An interprofessional quality improvement project to enhance delirium management and monitor delirium prevalence in a single PICU. *Pediatr Crit Care Med* 2017; 18:531–540
 49. Patel RV, Redivo J, Nelliott A, et al: Early mobilization in a PICU: A qualitative sustainability analysis of PICU Up! *Pediatr Crit Care Med* 2021; 22:e233–e242
 50. Norrenberg M, Vincent JL: A profile of European intensive care unit physiotherapists. European Society of Intensive Care Medicine. *Intensive Care Med* 2000; 26:988–994
 51. Kudchadkar SR: PICU UPI: Trial of a Multicomponent Early Mobility Intervention for Critically Ill Children, 2021. Available at: <https://clinicaltrials.gov/ct2/show/NCT03860168>. Accessed February 25, 2021